

Jathush Vivekanandam

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EDUCATION

State University of New York at Buffalo

Master of Science, Genetics, Genomics, and Bioinformatics

August 2023 – May, 2025

3.4 GPA

- Relevant Coursework: Molecular Biology, Biochemistry, Stats for Bioinformatics, Machine Learning, Statistical Data Mining

Bachelor's of Science, Biochemistry

August 2019 - May, 2023

3.2 GPA

- Relevant Coursework: Chemistry and Organic Chemistry Lab, Genetics, Pharmacology, Drug Action, Bioinformatics and Computational Biology, Statistical Programming in R

SKILLS

- **Molecular & Cellular Biology:** Nucleosome reconstitution, EMSA, PAGE gels, recombinant protein assays, chromatin biology, cancer transcription factor regulation
- **Molecular Assays Familiarity:** qPCR, Western blot, immunostaining, ELISA (collaborations & training)
- **Programming Languages:** Python, Linux, R
- **Computational Tools:** Snakemake, Git, VS Code, Visual Studios, Quarto, Microsoft Excel
- **Concepts:** Computational Genomics, Machine Learning, Biostatistics, Data Visualization, Data Harmonization
- **Bioinformatics Tools:** Bash, Conda, BEDTools, MACS2, LiftOver, Deseq2, GREAT, scikit-learn, pandas, numpy
- **Languages:** English (Fluent), Tamil (Fluent), French (Intermediate)

PROJECTS

Master's Thesis Research – Dr. Michael Buck Lab, SUNY Buffalo

Aug 2023 – May 2025

- Designed and performed nucleosome reconstitution assays with histone octamers and Widom 601 sequences.
- Conducted electrophoretic mobility shift assays (EMSAs) using recombinant his₆-KLF4 protein, applying Cy5-tagged DNA and PAGE gels for quantification.
- Investigated oncogenic transcriptional regulation by validating KLF4 interactions with p53 transcription factor binding sites.
- Developed supervised machine learning models on chromatin datasets (RNA-seq, ATAC-seq, and MNase-seq) to predict transcription factor (KLF4) binding and uncover genomic determinants in pluripotency and chromatin accessibility.
- Developed reproducible workflows for data normalization, batch effect correction, and model performance evaluation across >700,000 genomic sites.
- Highlighted chromatin accessibility and nucleosome architecture as key determinants of pioneer factor activity in cancer systems.

Graduate Researcher – Dr. Michael Buck Lab

Aug 2023 –

May 2025

- Developed machine learning models to predict transcription factor (KLF4) binding in pluripotency and oncology models.
- Built reproducible workflows for QC, normalization, and batch effect correction of NGS datasets.
- Generated visualizations and biomarker reports to guide experimental collaborators.

NIH R25 Scholar

May – Aug 2023

- Investigated FOXA1 mutant binding in breast and prostate cancer via motif and chromatin accessibility analysis.
- Presented findings at NIH symposia, bridging computational and molecular oncology.

Undergraduate Researcher – Dr. Michael Buck Lab

Dec 2021 – May 2023

- Conducted motif enrichment analysis on FOXA1 mutants in breast and prostate cancer models.
- Used BEDTools, MEME Suite, and Linux workflows to compare wild-type vs. mutant binding profiles.
- Produced publication-ready figures and collaborated with cancer biologists for validation.

POSTERS, TEACHING & LEADERSHIP

Bioinformatics Teaching Assistant

January 2025 – May 2025

- Supported students in statistical analysis and bioinformatics workflows for sequencing data.

Biochemistry Day Poster**May, 2023**

- Presented the pipeline and results for motif abundance analysis in ChIP-seq data of KLF4 in cancer and prostate cancer models.

Sri Lankan American Association of Rochester NY**May 2021 – Present***Youth Committee Director*

- Mentored youth in the Sri Lankan community of Rochester, NY in academic preparation and cultural education.